

1. Confidence Intervals (CI)

Definition and Purpose

A **confidence interval** is a range of values, derived from sample data, that is used to estimate an unknown **population parameter** (e.g., population mean, proportion). It gives a sense of the **uncertainty in an estimate**. The interval is constructed so that a specified proportion of such intervals from repeated samples would contain the true parameter.

Mathematically for a mean:

$$CI = \bar{x} \pm (\text{Critical value}) \times \frac{s}{\sqrt{n}}$$

where

- $\bar{x}$ = sample mean
- s = sample standard deviation
- n = sample size
- Critical value = value from z or t distribution based on confidence level

Confidence Level

The **confidence level** (e.g., 90%, 95%, 99%) represents the **probability** that the confidence interval procedure will capture the true parameter if we repeated sampling many times. A 95% CI implies that **95 out of 100** such intervals would contain the true value.

For example, if the sample mean of time spent on an app is 40 minutes with CI [36, 44] at 95% confidence, it means we are “95% confident” the true population mean lies within that range **based on our sample**.

Interpreting Confidence Intervals

- Correct interpretation: If we could repeat the study many times, **approximately 95% of computed intervals** would contain the true parameter. It **does not mean the parameter is 95% likely** to be in the specific interval you computed — the parameter is fixed.
- A **narrower interval** means more precision (larger sample or smaller variation).

- A **wider interval** indicates more uncertainty.
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2. Hypothesis Testing

Hypothesis testing is a **formal decision-making process** using sample data to evaluate a claim about a population parameter. It answers: "Is the observed data consistent with a given assumption (null hypothesis)?"

Key Components

a. Null Hypothesis (H_0)

The **starting assumption** or statement of **no effect or no difference**. It represents the default belief.

Example:

$$H_0: \mu = 50$$

This states that the average visits = 50.

b. Alternative Hypothesis (H_1 or H_a)

The opposite of H_0 , asserting that there *is a difference, effect, or change*.

Example:

$$H_1: \mu \neq 50$$

This says the average visits is not 50.

P-value

The **probability of observing the test statistic (or more extreme)** assuming the null hypothesis is true. Lower p-values provide stronger evidence *against* H_0 .

Significance Level (α)

A threshold set before testing. Common values are **0.05** (5%), **0.01**, or **0.10**. It represents the maximum allowable probability of making a **Type I error** (incorrectly rejecting a true null hypothesis).

Decision Rule Using P-value

- If **p-value $\leq \alpha$** $\rightarrow$ reject H_0 (results are statistically significant).
 - If **p-value $> \alpha$** $\rightarrow$ fail to reject H_0 (insufficient evidence).
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Types of Errors

- **Type I Error (False Positive)**: Reject H_0 when it is true; probability = α .
 - **Type II Error (False Negative)**: Fail to reject H_0 when it is false; probability = β .
 - **Power of a test** = $1 - \beta$; probability of correctly rejecting a false H_0 .
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3. Steps in Hypothesis Testing

1. **State H_0 and H_1** based on the research question.
 2. Choose **significance level α** (commonly 0.05).
 3. Select the **appropriate test statistic** (z, t, chi-square, F).
 4. Compute the **test statistic** from sample data.
 5. Determine the **critical value(s)** from the distribution (normal, t, chi-square, F).
 6. Compare test statistic with critical value or compare **p-value with α** .
 7. **Conclude** by rejecting or failing to reject H_0 .
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4. Common Hypothesis Tests with Examples and Mathematics

Z-Test

Used when the population standard deviation is **known** or sample size is large ($n > 30$).

Test statistic:

$$z = \frac{\bar{x} - \mu_0}{\sigma / \sqrt{n}}$$

Example:

Sample mean = 52, $\mu_0 = 50$, $\sigma = 10$, $n = 100$.

$$z = \frac{52 - 50}{10 / \sqrt{100}} = \frac{2}{1} = 2$$

At $\alpha = 0.05$ (two-tailed), critical $z \approx \pm 1.96$. Since $2 > 1.96$, reject H_0 .

T-Test

Used when **population standard deviation is unknown** and for small samples. Test statistic follows the **Student's t distribution** with special degrees of freedom.

Test statistic:

$$t = \frac{\bar{x} - \mu_0}{s / \sqrt{n}}$$

Example:

Sample mean = 75, $\mu_0 = 70$, $s = 8$, $n = 16$.

$$t = \frac{75 - 70}{8 / \sqrt{16}} = \frac{5}{2} = 2.5$$

Compare with t critical for $df = 15$ at $\alpha = 0.05$.

Chi-Square Test

Used for **categorical variables** to test association or goodness-of-fit; test statistic follows a χ^2 distribution.

Test statistic:

$$\chi^2 = \sum \frac{(O - E)^2}{E}$$

Where O = observed counts, E = expected counts.

ANOVA (Analysis of Variance)

Used to compare **means across three or more groups**. It tests whether at least one group mean is different. It uses the **F-statistic**. The analysis partitions total variability into:

- **SSB (Between Groups)**
- **SSW (Within Groups)**

F statistic:

$$F = \frac{MSB}{MSW} = \frac{SSB/df_B}{SSW/df_W}$$

Where df_B and df_W are degrees of freedom for between and within groups.

Example calculations:

Assume three groups with means and variances; compute SSB, SSW, then F.

If computed $F > \text{critical } F$ at (df_B, df_W), reject H_0 . (Group means are not all equal).

5. Critical Value Method

Instead of p-values, you can use **critical values**:

- Find critical value from distribution table based on α .
- Compare test statistic with critical value:
 - If test statistic is in the **rejection region**, reject H_0 .
 - Otherwise, fail to reject H_0 .

6. One-Tailed vs Two-Tailed Tests

- **Two-tailed test:** H_1 : parameter $\neq$ value; rejection areas on both tails of distribution.
 - **One-tailed test:** H_1 : parameter $>$ value or $<$ value; rejection on one tail only. The direction is based on the research question.
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7. Real-Life Example (Hypothesis Testing)

Suppose a company claims average customer waiting time is **5 minutes** ($H_0: \mu = 5$). You collect a sample of 50 customers with mean 5.6 and sample SD 1.2.

Compute t statistic:

$$t = \frac{5.6 - 5}{1.2/\sqrt{50}} \approx \frac{0.6}{0.1697} \approx 3.54$$

Compare with t critical for $df = 49$ at $\alpha = 0.05$ (≈ 2.009). Since $3.54 > 2.009$, reject H_0 — evidence that average waiting time differs from 5 minutes.

P-value corresponding to $t = 3.54$ would be < 0.01 ; less than $\alpha = 0.05$ again leads to rejecting H_0 .

8. Relationship Between Confidence Intervals and Hypothesis Testing

- A $1-\alpha$ **confidence interval** corresponds to a hypothesis test at significance level α .
- If the null hypothesis value (e.g., μ_0) **lies outside** the confidence interval, then H_0 is rejected.